Filter Order Optimization

Compound filter example

```
SELECT *
FROM Person
WHERE name IN ('Greiner', 'Halverhout') AND
      zipcode = 77055 AND
      birthdate > '1967-02-12';
```

Test	CPU cost/tuple	Selectivity
name IN	100ns	0.00001
zipcode =	1ns	0.0001
birthdate >	1000ns	0.3

400,000,000 tuples

Which order is best?

For each Person tuple:
- If name IN …
- Then if zipcode = …
- Then if birthdate > …
- Then output tuple

How many possible orders?
How to calculate cost for each?

Costs of two possible orders

400,000,000 tuples

Predicate	CPU cost/tuple	Selectivity	Cost for this predicate	Cost simplified
name IN	100ns	0.00001	400,000,000 × 100ns	40,000,000,000ns
zipcode =	1ns	0.0001	400,000,000 × 0.00001 × 1ns	4,000ns
birthdate >	1000ns	0.3	400,000,000 × 0.00001 × 0.0001 × 1000ns	400ns
				40,000,004,400ns

Predicate	CPU cost/tuple	Selectivity	Cost for this predicate	Cost simplified
zipcode =	1ns	0.0001	400,000,000 × 1ns	400,000,000ns
name IN	100ns	0.00001	400,000,000 × 0.0001 × 100ns	400,000ns
birthdate >	1000ns	0.3	400,000,000 × 0.00001 × 0.0001 × 1000ns	400ns
				400,400,400ns

Best ordering

Do we have to try all possible orders?

No!

Best: decreasing order $\frac{1-s(pred)}{cost(pred)}$

i.e., the fraction of data eliminated per unit time.